

Topic P3

Waves

Concept guide · Worked examples · Cambridge-style practice · Fully explained answers

What this guide teaches

A complete explanation of the syllabus concepts, common misconceptions, worked applications and exam-style practice for this topic.

Syllabus map

P3.1	General properties of waves
P3.2	Light
P3.3	Electromagnetic spectrum
P3.4	Sound

How to study: retrieve definitions and equations, cover the worked examples, then complete the practice paper without using the notes.

1. Wave quantities

A wave transfers energy without a net transfer of matter. Amplitude is maximum displacement from equilibrium; wavelength is distance between corresponding points; frequency is oscillations per second; period is time for one oscillation.

$$v = f\lambda \quad T = 1/f$$

Worked example

A water wave has frequency 5.0 Hz and wavelength 0.24 m. Speed = $5.0 \times 0.24 = 1.2$ m/s.

Examiner insight

Do not confuse wave speed with the speed at which particles of the medium oscillate.

2. Transverse and longitudinal waves

In a transverse wave, oscillations are perpendicular to energy transfer; examples include electromagnetic waves and waves on a stretched string. In a longitudinal wave, oscillations are parallel to energy transfer and produce compressions and rarefactions; sound in air is longitudinal.

Examiner insight

Use the direction of oscillation relative to travel, not the appearance of a diagram, to classify the wave.

3. Reflection, refraction and diffraction

Reflection changes direction at a boundary and obeys angle of incidence = angle of reflection. Refraction occurs because wave speed changes on entering a different medium; frequency stays constant, so wavelength changes. Diffraction is spreading at an edge or through a gap and is greatest when gap size is comparable to wavelength.

Worked example

A wave slows from 12 m/s to 8.0 m/s while frequency remains 4.0 Hz. Its wavelength changes from 3.0 m to 2.0 m.

Examiner insight

For refraction, state the speed change before stating whether the ray bends towards or away from the normal.

4. Light and lenses

A converging lens refracts parallel rays towards the principal focus. A real image can be projected on a screen; a virtual image cannot. Ray diagrams use a ray parallel to the axis that passes through the focus and a ray through the optical centre that continues undeviated.

magnification = image height / object height

Examiner insight

Measure angles from the normal, never from the surface.

5. Electromagnetic spectrum

In order of increasing frequency: radio, microwave, infrared, visible, ultraviolet, X-ray, gamma. All travel at 3.0×10^8 m/s in vacuum and are transverse. Higher frequency means shorter wavelength and greater photon energy. Uses and hazards depend on interaction with matter.

Examiner insight

Do not state that different electromagnetic regions have different speeds in a vacuum.

6. Sound

Sound is produced by vibrating sources and requires a medium. Frequency determines pitch; amplitude determines loudness. Humans typically hear about 20 Hz to 20 kHz. Echo timing can determine distance, and ultrasound has frequencies above 20 kHz.

distance = speed $\times$ time / 2

Worked example

An echo returns after 0.40 s when sound speed is 340 m/s. Distance to the wall = $340 \times 0.40 / 2$ = 68 m.

Examiner insight

Divide echo distance by two because the measured time includes the outward and return journeys.

Exam technique

High-value habits

Use precise definitions, show every calculation step, include units, apply concepts to unfamiliar situations, and answer the command word directly.

Retrieval checklist

- Recall all definitions and equations without notes.
- Explain one unfamiliar application using a linked chain of reasoning.
- Identify the first error in each incorrect response.
- Complete one timed set and mark it only after finishing.

Practice Paper

Topic P3

aves**Instructions**

Answer all questions. Select one option for multiple choice. Show working in open spaces and write explanations on the lines. Individual parts carry 1-3 marks.

Candidate name:

Class / date:

Total marks: 50

Suggested time: 60 minutes.

Section A · Multiple choice

Choose the best answer for each question.

1. Which quantity is measured in hertz? [1]
 - A frequency
 - B wavelength
 - C amplitude
 - D wave speed

2. A wave has frequency 4.0 Hz and wavelength 0.50 m. What is its speed? [1]
 - A 0.125 m/s
 - B 2.0 m/s
 - C 4.5 m/s
 - D 8.0 m/s

3. Which wave is longitudinal? [1]
 - A light in vacuum
 - B sound in air
 - C radio wave
 - D wave on a string

4. When light reflects from a mirror, the angle of reflection is measured from the [1]
 - A mirror surface
 - B incident ray
 - C normal
 - D reflected ray

5. Light enters glass from air and slows down. It bends [1]
 - A away from the normal
 - B towards the normal
 - C along the boundary
 - D without changing direction

6. When does the greatest diffraction occur? [1]
 - A gap much wider than wavelength
 - B gap similar to wavelength
 - C frequency is zero
 - D amplitude is small

7. Which electromagnetic wave has the highest frequency? [1]
 - A radio
 - B infrared
 - C X-ray
 - D gamma

8. Which statement about electromagnetic waves in vacuum is correct? [1]
- A They all have the same speed.
 - B Only visible light transfers energy.
 - C They are longitudinal.
 - D Radio waves travel slowest.
9. What determines the pitch of a sound? [1]
- A amplitude
 - B frequency
 - C speed only
 - D echo time
10. An echo returns in 0.20 s. Sound speed is 340 m/s. How far away is the wall? [1]
- A 17 m
 - B 34 m
 - C 68 m
 - D 1700 m

Section B · Structured questions

11 A ripple tank produces waves of frequency 6.0 Hz and wavelength 0.080 m.

(a) Calculate the wave speed. [2]

(b) The frequency doubles while speed is unchanged. Calculate the new wavelength. [2]

(c) State what is transferred by the wave and what is not transferred overall. [2]

(d) Define amplitude. [2]

[Total: 8]

12 A ray of light travels from air into glass.

(a) State what happens to its speed. [1]

(b) State whether it bends towards or away from the normal. [1]

(c) Explain why its wavelength changes but its frequency does not. [3]

(d) State three rules or construction lines used in a converging-lens ray diagram. [3]

[Total: 8]

13 A student investigates diffraction using water waves.

(a) State what is meant by diffraction. [2]

(b) Describe how the amount of diffraction changes as the gap becomes narrower. [2]

(c) State the condition for maximum diffraction. [2]

(d) Explain why radio waves diffract around buildings more than visible light. [2]

[Total: 8]

14 Electromagnetic waves are used in communication and medicine.

(a) Place infrared, ultraviolet and microwaves in order of increasing frequency. [2]

(b) State one use of microwaves and one associated hazard. [2]

(c) Explain why gamma radiation can damage living tissue. [2]

(d) Calculate the wavelength of a 1.0×10^6 Hz radio wave in vacuum. [2]

[Total: 8]

15 A student stands in front of a wall and hears an echo 0.50 s after clapping. The speed of sound is 340 m/s.

(a) Calculate the distance from the student to the wall. [3]

(b) Explain why the time must be divided by two. [1]

(c) State how frequency and amplitude affect the sound heard. [2]

(d) Explain why sound cannot travel through a vacuum. [2]

[Total: 8]

Worked solutions

Remove or fold back this section while completing the practice paper.

Marking routine

Award marks for correct physics and method, not only the final value. Mark each part separately.

Solutions · Multiple choice

1 A

2 B

3 B

4 C

5 B

6 B

7 D

8 A

9 B

10 B

Solutions · Structured questions

11. 11(a) $v=f\lambda=6.0\times0.080=0.48$ m/s [2]. (b) $\lambda=v/f=0.48/12=0.040$ m [2]. (c) Energy is transferred [1], with no net transfer of matter [1]. (d) Maximum displacement from equilibrium [2].

12. 12(a) Speed decreases [1]. (b) Towards the normal [1]. (c) Frequency is fixed by the source [1]; speed decreases [1]; therefore $\lambda=v/f$ decreases [1]. (d) Any three correct standard rays/constructions [1 each].

13. 13(a) Spreading of waves at an edge or through a gap [2]. (b) Diffraction increases as the gap narrows [2]. (c) Gap width comparable to wavelength [2]. (d) Radio wavelength is much longer and therefore more comparable to building dimensions [2].

14. 14(a) microwaves, infrared, ultraviolet [2]. (b) Valid use and linked heating hazard [2]. (c) It ionises atoms / damages cells or DNA [2]. (d) $\lambda=c/f=3.0\times10^8/1.0\times10^8=3.0$ m [2].

15. 15(a) $d=vt/2=340\times0.50/2=85$ m [3]. (b) Sound travels to the wall and back [1]. (c) Frequency controls pitch; amplitude controls loudness [2]. (d) There are no particles to pass on the vibration [2].

Self-assessment

Circle the first incorrect step and classify it as knowledge, equation choice, rearrangement, unit conversion, graph/diagram interpretation or communication.

Evidence used: Cambridge IGCSE Co-ordinated Sciences 0654 syllabus 2025-2027 and June 2026 papers, mark schemes and Principal Examiner Report supplied by the user. All practice questions are newly written.